Kristin Henderson

Leesburg, VA • kdhenderson@gmail.com • https://kdhenderson.github.io
https://github.com/kdhenderson • https://www.linkedin.com/in/kdhende

TECHNICAL SKILLS

- **Programming:** Python, R, SAS, MATLAB, SQL, Bash
- **Data & Analysis Libraries:** NumPy, pandas, dplyr, tidyverse, scikit-learn, XGBoost, PyTorch
- **Machine Learning & AI:** RAG, FAISS, agentic workflows, prompt engineering, LoRA fine-tuning, contrastive learning, transfer learning, Hugging Face, LangChain
- **Data Visualization:** matplotlib, seaborn, ggplot2, D3.js, Tableau, R Shiny
- **Data Formats & Infrastructure:** Parquet, PostgreSQL, MySQL, AWS (RDS, EC2), LSF clusters
- **Documentation & Version Control:** Git, GitHub, Jupyter notebooks, Quarto, Markdown

SELECTED PROJECTS

- Built a decision analysis framework in R to score the investment opportunity of emerging brands for a national retail brokerage / distribution firm, processing a large proprietary Parquet dataset with dplyr and benchmarking rankings with ordinal logistic regression.
- Developed a Retrieval-Augmented Generation (RAG) pipeline in Python with contrastive-trained embedding models, a LoRA fine-tuned language model, and cross-encoder reranking.
- Built an agentic workflow using LangChain and OpenAI SDK that routes queries to specialized RAG agents, retrieves context, and generates formatted responses.
- Built a graph-based system using NetworkX to analyze migration and species co-occurrence.
- Built Python scripts to download and consolidate electron microscopy image datasets and metadata from multiple public repositories for ML pipeline development.
- Applied time-series forecasting methods (ARMA, VAR, MLP, ensemble) in R to model multi-horizon river flow rates for resource planning.
- Authored an open-source Quarto-based statistics reference linking theory to applied data science: kdhenderson.github.io/stats-notes-for-ds

WORK EXPERIENCE

Howard Hughes Medical Institute, Janelia Research Campus, Ashburn, VA

Research Technician III, Jayaraman Lab. Dec 2021 – Present

- Built semi-automated Python pipelines to collect and process 100+ GB/day of behavioral video data for downstream statistical analysis and modeling.
- Applied statistical analysis and machine learning methods to identify patterns and behavioral structure from video tracking data.
- Collaborated with scientists and engineers to define analytical questions, assess data availability, and develop quantitative analyses and visualizations.
- Documented hardware, software, and analysis workflows to support data quality, reproducibility, and cross-team use.

Research Technician II, Keleman Lab. Jan 2017 – Dec 2021

- Led data analysis and experiments studying learning and memory; prepared datasets, figures, and results for publication.

Research Technician I, Project Technical Resources. June 2015 – Dec 2016

- Supported high-precision microscopy data collection for downstream quantitative analyses.

EDUCATION

Master of Science, Data Science (May 2026), Southern Methodist University, Dallas, TX
Bachelor of Science, emphasis in Biology, The Evergreen State College, Olympia, WA